

1 Proposed MS Thesis Topic

1.1 Working Title

Intervention-Aware Spatiotemporal Point Process Modeling for Predictive Multi-Robot Bird Deterrence in Vineyards

1.2 Why This Topic Fits the Existing Framework

The current implementation already provides:

- Health-weighted zone partitioning via a power diagram (UGV zones) with neighbor graph extraction.
- A decentralized online spatiotemporal intensity estimator per robot (OnlineSESTPP) with self-excitation and cross-excitation near zone borders.
- A task pipeline where detections generate immediate deterrence tasks and the predicted intensity field generates patrol tasks via hotspot extraction.
- A capability-aware task assigner that selects primary and secondary robots based on ETA, endurance, zone membership, and health.

However, deterrence actions currently do not feed back into the event intensity model. This thesis makes that feedback explicit by modeling deterrence as an *intervention* that suppresses future event intensity, enabling predictive planning that is aware of how actions change future risk.

1.3 Core Research Question

How can we design an online spatiotemporal point process model that (i) predicts bird intrusion hotspots from distributed detections and (ii) explicitly represents the suppressive effect of deterrence actions, so that multi-robot task selection reduces future bird activity and value-weighted exposure more effectively than reactive or prediction-only baselines?

1.4 Thesis Objectives

1. Develop an **intervention-aware** online spatiotemporal intensity model by extending the current self-exciting formulation with an inhibitory term driven by deterrence actions.
2. Integrate the new model into the existing task pipeline by scoring candidate patrol and deterrence locations using **expected future intensity reduction** per unit cost.
3. Evaluate performance in simulation against strong baselines (reactive and prediction-only) using metrics aligned with agricultural impact, energy, and coordination overhead.

1.5 Problem Setting and Notation

Let $\Omega \subset \mathbb{R}^2$ denote the vineyard area (or a discretized grid over the area). Robots are indexed by $r \in \mathcal{R}$ and each robot maintains a local model over its zone Ω_r .

Events (bird detections): $\mathcal{E}_r = \{(x_i, t_i)\}$ are detections attributed to robot r (plus cross-shared neighbor events).

Interventions (deterrence actions): $\mathcal{U}_r = \{(z_j, \tau_j, m_j)\}$ are completed deterrence actions executed by robot r at location z_j and time τ_j with deterrence mode or formation label m_j .

1.6 Baseline Model: Online Self-Exciting Spatiotemporal Intensity

The current OnlineSESTPP structure corresponds to a Hawkes-type intensity:

$$\lambda_r(x, t) = \mu_r(x) s(t) + \sum_{(x_i, t_i) \in \mathcal{E}_r} \alpha_{\text{in}} K_\sigma(x - x_i) \exp\left(-\frac{t - t_i}{\omega}\right) + \sum_{(x_k, t_k) \in \mathcal{E}_{\mathcal{N}(r)}} \alpha_{\text{cross}} w_{r \leftarrow k} K_\sigma(x - x_k) \exp\left(-\frac{t - t_k}{\omega}\right), \quad (1)$$

where $\mu_r(x)$ is a background rate, $s(t)$ is a diurnal multiplier, and $K_\sigma(\cdot)$ is a spatial Gaussian kernel.

1.7 Proposed Extension: Intervention-Aware Self-Exciting and Self-Regulating Model

We introduce an inhibitory (self-regulating) component driven by deterrence actions:

$$\lambda_r^{\text{IA}}(x, t) = \left[\lambda_r(x, t) - \underbrace{\sum_{(z_j, \tau_j, m_j) \in \mathcal{U}_r} \beta(m_j, x, \tau_j) K_{\sigma_u(m_j)}(x - z_j) \exp\left(-\frac{t - \tau_j}{\omega_u(m_j)}\right)}_{\text{intervention suppression}} \right]_+, \quad (2)$$

where $[\cdot]_+ = \max(\cdot, 0)$ enforces nonnegativity of intensity.

Interpretation:

- $\beta(\cdot)$ controls the immediate strength of deterrence.
- $\omega_u(\cdot)$ controls how long deterrence effects persist.
- $\sigma_u(\cdot)$ controls the spatial footprint of deterrence, which can encode formation footprint (for example, a wider formation yields larger σ_u).

1.7.1 Optional Habituation Model (Diminishing Returns)

To represent habituation, define a local habituation state $H(x, t)$ that increases with repeated interventions and decays over time:

$$H(x, t + \Delta t) = \rho H(x, t) + \sum_{(z_j, \tau_j, m_j) \in \mathcal{U}(t, t + \Delta t)} K_{\sigma_H}(x - z_j), \quad 0 < \rho < 1. \quad (3)$$

Then deterrence strength decreases with habituation:

$$\beta(m, x, t) = \beta_0(m) \exp(-k_H H(x, t)). \quad (4)$$

This connects directly to the agricultural motivation: repeating the same deterrence pattern in the same region yields weaker future benefit.

1.8 Planning: Scoring Actions by Expected Future Reduction

At each replan cycle, generate candidate targets from the current predicted intensity field (hotspots), then score each candidate deterrence action by its expected reduction in future intensity over a horizon T_H .

Let a candidate action be $a = (z, t, m)$ executed at current time t . Define the reduction proxy:

$$\Delta J(a) = \int_t^{t+T_H} \int_{\Omega_r} w(x) \left(\lambda_r^{\text{IA}}(x, \tau) - \lambda_r^{\text{IA},a}(x, \tau) \right) dx d\tau - c_E(a), \quad (5)$$

where $w(x)$ is the value weighting over the vineyard (premium rows, edges, phenology), and $c_E(a)$ is an energy or time cost.

A computationally efficient approximation follows from the exponential temporal kernel:

$$\int_t^{t+T_H} \exp\left(-\frac{\tau-t}{\omega_u}\right) d\tau = \omega_u \left(1 - \exp\left(-\frac{T_H}{\omega_u}\right) \right). \quad (6)$$

This allows scoring many candidates with low overhead on a grid.

1.9 Decentralized Neighbor Sharing With Interventions

The existing boundary-event mechanism can be extended to share *intervention events* near zone boundaries in addition to detection events:

- If a deterrence action completes within a distance d_b of the zone boundary, broadcast a compact message $(z, \tau, m, \beta, \sigma_u, \omega_u)$ to neighbor zones.
- Neighbors add that event into their local inhibitory term (cross-inhibition), improving prediction consistency near borders.

This remains communication-efficient because it is event-triggered and local.

1.10 Implementation Plan Within the Existing Codebase

1.10.1 Model Extension (SESTPP.py)

Add a second grid accumulator for inhibition mass, analogous to trigger mass:

- `inhib_mass[y,x]` updated by `add_intervention_event()` with spatial stamping similar to `_stamp()`.

- In `advance_time(dt)`, decay `trigger_mass` with $\exp(-dt/\omega)$ and decay `inhib_mass` with $\exp(-dt/\omega_u)$ (possibly mode dependent).
- Update intensity as `lam = clip(mu*time_mult + trigger_mass - inhib_mass)`.

1.10.2 Closed-Loop Integration (DeterrentSystem.py)

When a deterring task completes, call:

- `robots[rid].m.add_intervention_event(x,y,mode=m)` at completion time.
- Broadcast intervention boundary events to neighbors when near the boundary (reuse Event-Bus).

1.10.3 Action Scoring (TaskGenerator.py)

Replace or augment the patrol target selection:

- Generate candidate points from `hotspots()` (already implemented).
- For each candidate, compute $\Delta J(a)$ using the approximation in (6).
- Add the best scoring patrol or deterrence task per robot per cycle (respecting cooldown).

1.11 Experimental Design

1.11.1 Simulation Scenarios

Use a ground-truth generative process for bird events so interventions can have measurable impact:

- Base rate plus self-excitation to generate clustered bird activity.
- Optional edge preference and row value map for $w(x)$.
- Detection noise model (false negatives) consistent with sensor realism.
- Intervention effect applied to the ground-truth intensity to represent actual deterrence impact.

1.11.2 Baselines

At minimum:

1. Reactive: respond only to detections, no predictive patrol.
2. Prediction-only: current self-exciting model generates hotspots but deterrence does not suppress predicted risk.
3. Proposed: intervention-aware self-exciting and self-regulating model with action scoring by expected reduction.

Optional ablations:

- No neighbor sharing vs neighbor sharing of detections only vs detections plus interventions.
- No habituation vs habituation-aware suppression (4).

1.11.3 Metrics

Report mean and variance over multiple randomized runs:

- Value-weighted exposure:

$$J_{\text{exp}} = \int_0^T \int_{\Omega} w(x) \lambda^{\text{true}}(x, t) dx dt.$$

- Mean response time from event onset to first deterrence arrival.
- Total energy or travel distance per robot (UGV and UAV separated).
- Task efficiency: completed tasks per unit distance, or reduction in exposure per task.
- Communication overhead: number of boundary messages and total bytes sent (proxy).

1.12 Expected Contributions

- A practical online intervention-aware spatiotemporal point process model suitable for real-time robotics.
- A planning objective and scoring method that selects actions based on predicted future reduction, not only current hotspots.
- A decentralized coordination mechanism that shares both detections and intervention effects locally to reduce border artifacts.
- An evaluation showing measurable reduction in value-weighted exposure and improved responsiveness under realistic constraints.

1.13 Minimum Deliverables and Stretch Goals

Minimum deliverables:

- Implement (2) and integrate it into the simulation loop.
- Implement action scoring (5) and compare against baselines.
- Provide a full experimental section with metrics and ablations.

Stretch goals (if time permits):

- Habituation model (3) and (4) with mode-dependent parameters.
- Online parameter adaptation for $\beta_0(m)$ and $\omega_u(m)$ using likelihood-based updates.
- Formation-dependent footprints by mapping formation choice to $\sigma_u(m)$ and comparing footprints empirically.

1.14 Proposed Thesis Chapter Outline

1. Introduction and problem motivation
2. Related work: persistent monitoring, hotspot prediction, spatiotemporal point processes, intervention modeling
3. System overview and existing architecture
4. Intervention-aware point process model (formulation and online update)
5. Planning and task selection using expected future reduction
6. Experimental setup and baselines
7. Results, ablations, and discussion
8. Conclusions and future work